Sahil Verma

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Research Interests

My research is broadly focused on Trustworthy ML specifically focused on robustness and interpretability in ML. I am currently focusing on AI Safety Research.

Education

Sept 2019 - Present	PhD in Computer Science **University of Washington, Seattle** Advisors: Jeff Bilmes and Chirag Shah
July 2015 - July 2019	BTech in Electrical Engineering **Indian Institute of Technology Kanpur (IIT Kanpur)** Advisor: Subhajit Roy

Honors and Awards

2024	Conference Travel Grant of $1000	SATML 2024
2023	Best Oral Presentation Award	BUGS Workshop at NeurIPS
2022	Best Student Paper Award	TEA Workshop at NeurIPS
2020	Best Paper Award and Nvidia Titan RTX GPU	ML-RSA Workshop at NeurIPS
2019	Allen School Fellowship	Paul G. Allen School, UW
2017	Student Travel Award of $1800 for FSE	Google India
2015	All India Rank 663	IITJEE Advanced Exam

Publications

OmniGuard: An Efficient Approach for AI Safety Moderation Across Modalities
Sahil Verma, Keegan Hines, Jeff Bilmes, Charlotte Siska,
Luke Zettlemoyer, Hila Gonen, Chandan Singh
under review

How Many Van Goghs does it take to Van Gogh?
Sahil Verma, Royi Rassin, Arnav Mohanty Das, Gantavya Bhatt, Preethi Seshadri,
Chirag Shah, Jeff Bilmes, Hannaneh Hajishirzi, Yanai Elazar
ATTRIB, RegML, and SafeGENAI at NeurIPS 2024 and NLLP at EMNLP 2024

Effective Backdoor Mitigation Depends on the Pre-training Objective
Sahil Verma, Gantavya Bhatt, Soumye Singhal, Arnav Das, Chirag Shah,
John P Dickerson, Jeff Bilmes
TMLR and BUGS at NeurIPS 2023 **(Best Oral Paper)**

RecRec: Algorithmic Recourse for Recommender Systems
Sahil Verma, Ashudeep Singh, Varich Boonsanong, John P Dickerson, Chirag Shah
CIKM 2023 (Short Paper)

Post-Hoc Attribute-Based Explanations for Recommender Systems
Sahil Verma, Chirag Shah, John P. Dickerson, Anurag Beniwal,
Narayanan Sadagopan, Arjun Seshadri
TEA Workshop at NeurIPS 2022 **(Best Student Paper Award)**

Amortized Generation of Sequential Counterfactual Explanations for Black-box Models
Sahil Verma, Keegan Hines, John P Dickerson
AAAI 2022

Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review
Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John P. Dickerson, Chirag Shah
CSUR Computing Surveys and ML-RSA Workshop at NeurIPS 2020 **(Best Paper Award)** **(800+ Citations)**

Removing biased data to improve fairness and accuracy
Sahil Verma, Michael Ernst, Rene Just

Fairness Definitions Explained
Sahil Verma and Julia Rubin
FairWare Workshop at ICSE 2017 **(1500+ Citations)**

Facets of Fairness in Search and Recommendations
Sahil Verma, Ruoyuan Gao, Chirag Shah
Algorithmic Bias Workshop at ECIR 2020

ShapeFlow: Dynamic Shape Interpreter for TensorFlow
Sahil Verma and Zhendong Su

Debug-Localize-Repair: A Symbiotic Construction for Heap Manipulations
Sahil Verma and Subhajit Roy
FMSD Journal 2021

Synergistic Debug-Repair for Heap Manipulations
Sahil Verma and Subhajit Roy
ESEC/FSE 2017

Patents

Amortized Generation of Sequential Counterfactual Explanations for Black-box Models
Sahil Verma, Keegan Hines, John P Dickerson
U.S. Patent Application No.: 17/520,069

Work Experience

June 2024 - Current	Research Intern at MSR, Redmond. Working on Jailbreak Defense with Chandan Singh.
June 2022 - Oct 2022	Research Intern at Amazon, USA. Developed novel explainability technique for recommender systems.
June 2020 - Sept 2021	Research Fellow at Arthur AI, USA. Developed industry deployable ML explainability algorithm.
June 2019 - Sept 2019	Research Intern at ETH Zurich, Switzerland. Developed tensor shape incompatibility bugs detection in TensorFlow.
May 2018 - Aug 2018	Research Intern at CSAIL, MIT, USA. Developed tool for automating floating bit allocation in programs.
May 2017 - Aug 2017	Research Intern at NUS, Singapore. Developed tool to convert CSP programs into C code.

Professional Responsibilities

- **Reviewed research papers for:**

- *Workshops*: AFCR 2021 (3), AFCP 2022 (7), HCAI 2022 (4), R0-FoMo NeurIPS 2023 (2), BUGS NeurIPS 2023 (2).
- *Conferences*: EAAMO 2021 (1), XAIF 2021 (3), NeurIPS 2022 (1), AAAI 2022 (1), ICML 2022 (1), FAccT 2022 (1), AIES 2022 (1), XAIF 2022 (2), AAAI 2023 (5), ECCV 2024 (3), EMNLP 2024 (1), ACL 2024 (1), ICLR 2025 (3).
- *Journals*: IEEE Transactions on Artificial Intelligence (1), Data Mining and Knowledge Discovery (1), International Journal of Data Science and Analytics (1), Journal of Decision Systems (1), Computer and Operations Research (1), Machine Learning (2), IEEE Transactions on Knowledge and Data Engineering (1)

- **Student Volunteer** at ESEC/FSE 2017.